WORK ENERGY & POWER

Work done by constant force:

FORCE

- How long (Δt): $\boxed{f \cdot \Delta t = \Delta P}$
- How far (S): $\boxed{F \cdot S = \text{Work} = \Delta K \cdot E}$

Work

- By constant force: Work $= \vec{f} \cdot \vec{s}$ $= FS\cos\theta$
- By variable force: $W = \int \vec{f} \cdot \vec{ds}$ $= \int f\,ds\cos\theta$

W → Area under f-S graph

F → slope of W-S graph

$$\boxed{W = \int_{x_1}^{x_2} f_x dx + \int_{y_1}^{y_2} f_y dy + \int_{z_1}^{z_2} f_z dz}$$

- $W_{total} = W_1 + W_2 + W_3 + \ldots$
  $= \vec{f_1} \cdot \vec{s_1} + \vec{f_2} \cdot \vec{s_2} + \vec{f_3} \vec{s_3} + \ldots$
- $W = \vec{f} \cdot \vec{s}$
  ↳ doesn't depend on nature of Motion.

#NOTE:- [Work]

- SI unit → Nm or Joule
- C.G.S unit → dyne cm
- Scaler quantity, $[ML^2T^{-2}]$
- May be +ve, −ve, zero.
- Depends upon frame.
- work done by One force doesn't depend on other force.

$\int W \cdot ds$ → NAALI ← W ; $\int f \cdot ds$ = Area

W → Force: $\frac{dw}{ds}$ = slope of W−S

Force → NAALI: df/dt

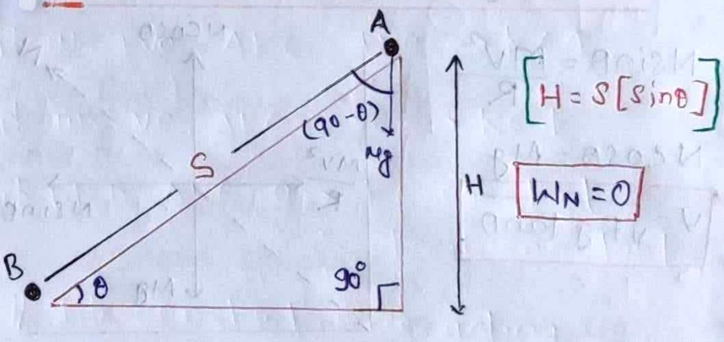
Work = +ve	Work = 0	Work = −ve
$0 \le \theta < 90^\circ$	$\theta = 90^\circ$	$90^\circ < \theta \le 180^\circ$
W = +ve	W = 0	W = −ve
K.E = ↑	KE = const^n	K.E ↓
	speed = const^n	speed ↓

Zero Work

- $\theta = 90^\circ$ (Angle b/w $\vec{F}$ & $\vec{S}$) [W = 0]
- Work done by normal = 0
- work done by centripetal force = 0

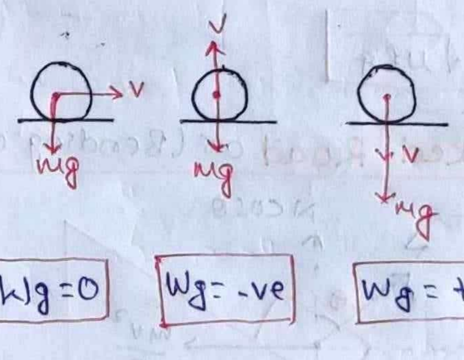
Work done by gravity

↳ does not depends on path



$$\boxed{W_g = MgH = MgS\sin\theta}$$

→



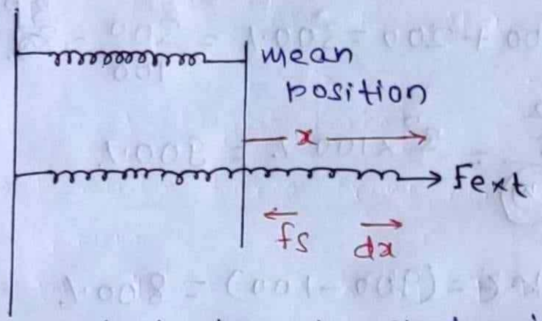
$\boxed{W_g = 0}$ $\boxed{W_g = -ve}$ $\boxed{W_g = +ve}$

Work done by friction

- Depends on path.
- Friction always acts along the path.

$$\boxed{W_{friction} = -f_r S}$$

Work done by spring force



★ Work done by spring in elongation/compression = $\boxed{-ve}$

↳ Work done in compression / Elong. from x_1 to x_2 :-

$dw = f_s dx \cos 180°$

$$\boxed{W = -\frac{1}{2}K\left[x_2^2 - x_1^2\right]}$$

- If $x_1 = 0$ (mean) to $x_2 = x$ (Elongation)

$$\boxed{W = -\frac{1}{2}Kx^2}$$

- compression from mean to x.

$$\boxed{W_{spring} = \frac{1}{2}Kx^2}$$

Momentum & Kinetic Energy

→ Momentum → vector ; → dirn along velocity

$$\boxed{\vec{P} = m\vec{V}}$$

K·E = (Energy stored due to motion)
↳ Scaler , V → speed

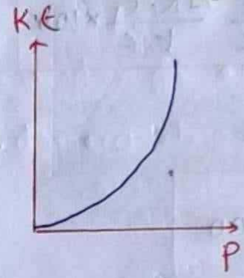
$$\boxed{K \cdot E = \frac{1}{2}mv^2 = \frac{P^2}{2m}}$$

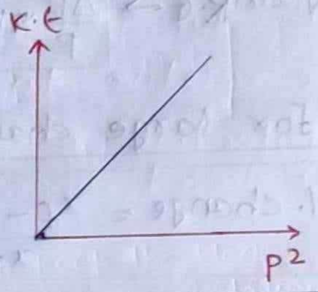
$$\boxed{P = \sqrt{2mK \cdot E}}$$

- Two identical particles having same KE may / may not have same Momentum.
- 2-Identical bodies having same momentum must have same K·E.

Graph:

$K \cdot E = \frac{P^2}{2m}$ (m = constant)



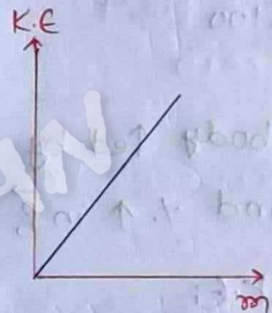


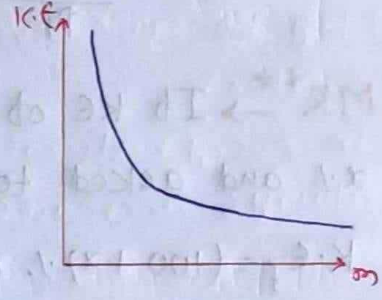
$K \cdot E = \frac{1}{2}m v^2$; v^2 → same

$\boxed{K \cdot E \propto m}$

$K \cdot E = \frac{P^2}{2m}$; P^2 → same

$\boxed{K \cdot E \propto \frac{1}{m}}$



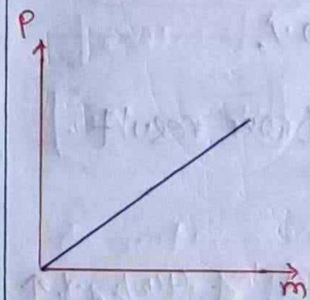


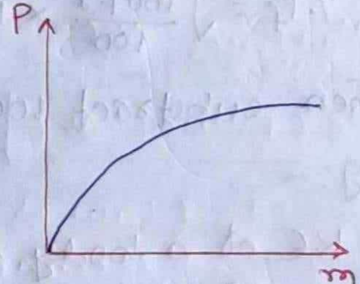
$\vec{P} = m\vec{V}$; $\vec{V}$ → Const

$P \propto m$

$P = \sqrt{2mK \cdot E}$; K·E → const.

$P \propto \sqrt{m}$





$P = \sqrt{2mK \cdot E}$

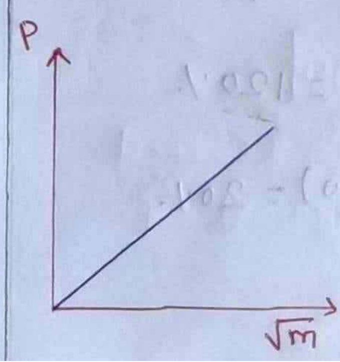
$K \cdot E = \text{const}$

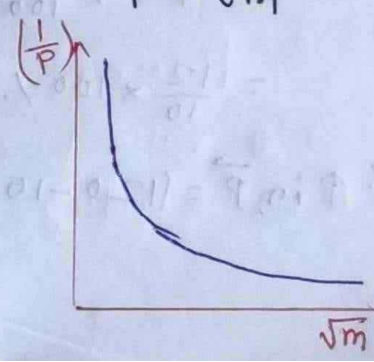
$P \propto \sqrt{m}$

$K \cdot E = \text{const}$

$P = \sqrt{m}$

$\frac{1}{P} = \frac{1}{\sqrt{m}}$





% Change

for small change ($\leq 5\%$)

→ Use Error Analysis:

If $m = \text{const}$

$K.E = \frac{p^2}{2m} \rightarrow \boxed{\frac{\Delta K.E}{K.E} \times 100 = 2\left[\frac{\Delta p}{p} \times 100\right]}$

$p = \sqrt{2m K.E} \rightarrow \boxed{\frac{\Delta p}{p} \times 100 = \frac{1}{2}\, \frac{\Delta K.E}{K.E} \times 100}$

for large change

$$\boxed{\%\ \text{change} = \frac{x_f - x_i}{x_i} \times 100}$$

- $\%\ \text{change} = \frac{KE_f - K.E_i}{K.E_i} \times 100$
- $\%\ \text{change} = \frac{P_f - P_i}{P_i} \times 100$

↳ MR** → If KE of a body ↑ed by $x\%$ and asked to find % ↑ in $\vec{P}$,

$K.E_f = (100 + x)\%$ of $K.E_i$

$P_f = \sqrt{K.E_f} = \sqrt{\frac{100+x}{100} \times K.E_i}$

$\%\ P_f = \sqrt{\frac{100+x}{100}} \times 100\%$

Then substract 100 from result.

E.g:-

① KE of a body ↑ed by 44%, % ↑ in $\vec{P}$?

→ $K.E_f = 144\%\ K.E_i$

$P_f = \sqrt{K.E_f} = \sqrt{\frac{144}{100} K.E_i}$

$= \frac{12}{10} \times 100\% = 120\%$

$\%\ \uparrow$ in $\vec{P} = (120 - 100) = 20\%$.

② K.E ↓ed by 19%, % ↓ in $\vec{P}$?

→ $K.E_f = 81\%$

$P_f = \sqrt{\frac{81}{100} K.E_i} = \frac{9}{10} \times 100\% = 90\%$

$\%\ \downarrow = 100 - 90 = 10\%$

③ $\vec{P}$ ↑ by 200%, KE ↑ by?

→ $P_f = 100 + 200 = 300\% = \frac{300}{100} = 3P_i$

$K.E_f = \frac{P_f^2}{\cancel{2m}} = 3^2 \times 100\% = 900\%$

$\%\ \uparrow$ in $K.E = (900 - 100) = 800\%$.

Work - Energy theorem ↳ Never fails

- Total work done by all forces = change in K.E

$$\boxed{W_{total} = \Delta KE}$$

$$\boxed{W = \vec{F} \cdot \vec{S} = \Delta KE}$$

if $W = +ve$, $\Delta KE = +ve$ → KE ↑

if $W = -ve$, $\Delta KE = -ve$ → KE ↓

if $W = 0$, $\Delta KE = 0$ → KE constant

∴ speed constant

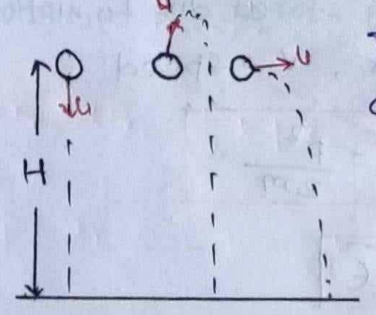
If constant force acting

$$\boxed{KE \propto v^2 \propto t^2}$$

→ $\boxed{\Delta KE = F.d}$

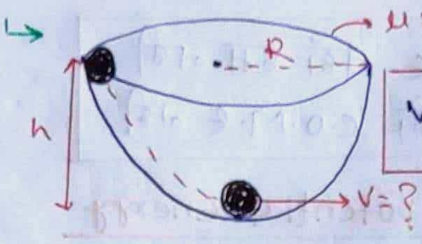
↳ doesn't depend on mass.

→

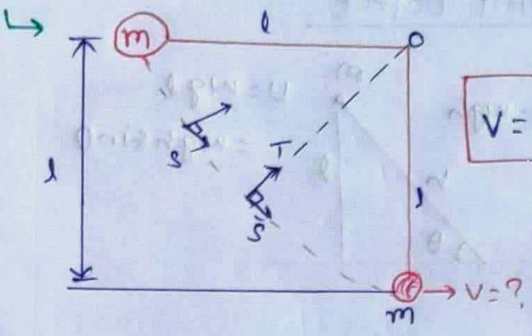


Their speed of collision with ground = same

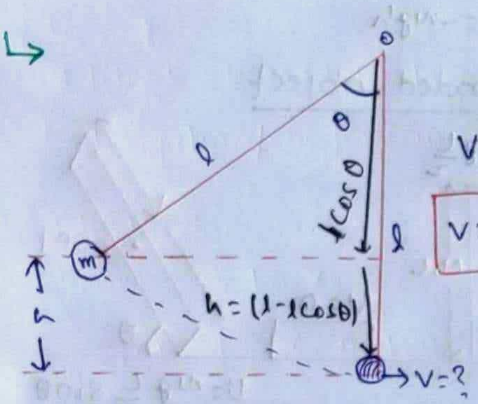
• Find → ?



$$\boxed{V=\sqrt{2g\times(\text{vertical height})}}$$

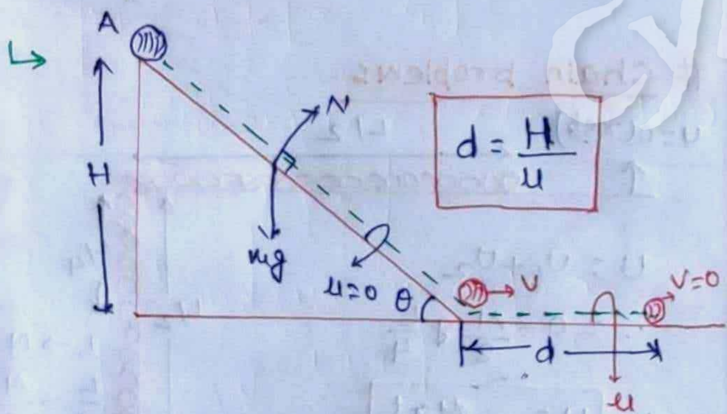


$$\boxed{V=\sqrt{2gl}}$$



$V=\sqrt{2gh}$

$$\boxed{V=\sqrt{2g(l-l\cos\theta)}}$$



$$\boxed{d=\frac{H}{\mu}}$$

Conservative force (C.f)

• Work done does not depend on path, only depends on initial & final position.

• Work done in closed path = 0

$$\boxed{W_{cf}=\oint \vec{F}\cdot d\vec{r}=0}$$

E.g → Gravitational force, electrostatic force spring force

• P.E stored due to −ve work done by C.f.

$\boxed{-W_{C.f}=\Delta U}$

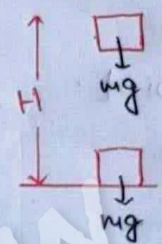
• Work done by C.f is reversible.

Non-conservative force (N.C.f)

• Work done depends on path.
• Potential Energy not defined for non-conservative force.
• Work done by N.C.F is non-reversible.
• Work done in close loop may or may not be zero

E.g → Friction, Normal, Tension

→ During downward Motion



$$\boxed{W_g=\Delta KE}$$

$$\boxed{PE\downarrow = KE\uparrow}$$

$\boxed{\Delta U=-W_{cf}}$

$\Rightarrow \Delta U=-\int \vec{f}_{c.f}\cdot d\vec{r}$

$\boxed{F_{cf}=-\left(\frac{dU}{dr}\right)}$ → PE gradient (vector)

$$\boxed{\vec{F}_{cf}=-\left(\frac{dU}{dx}\right)\hat{i}-\left(\frac{dU}{dy}\right)\hat{j}-\left(\frac{dU}{dz}\right)\hat{k}}$$

$$\boxed{\Delta U=-\int f_x dx-\int f_y dy-\int f_z dz}$$

$$\boxed{\Delta U=U_f-U_i=-W_{Cf}}$$

If $W_{cf}=-ve$

$\Delta U=+ve$

PE ↑

If $W_{cf}=+ve$

$\Delta U=-ve$

PE ↓

Change in Potential Energy (ΔU)

- It just a name given to -ve work of conservative force.
- P.E not defined or, it is not a physical quantity only is change in Potential energy is defined.
- Value of potential energy is not absolute (Unique) it relative & relative to frame of reference, Potential Energy depends on ref.
- ΔU doesn't depend on reference.

Potential Energy

- Energy stored due to shape, size & position.
- depends on reference point.
- May be +ve, -ve, zero.
- Scaler
- In every natural phenomena, P.E always decreases.

Work Energy Theorem

↳ Always valid

Father — Baap of all laws — Mother

$W_{\text{all forces}} = \Delta KE$

$\boxed{W_{C.f} + W_{N.C.f} = \Delta KE}$

$\boxed{W_{C.f} = -\Delta U}$

↳ Beta (Law of conservation of (C.O.M.E) Mechanical Energy)

↳ only valid when $Ncf = 0$

$\boxed{(K.E + U)_{initial} = [K.E + U]_{final}}$

↳ Beti $\boxed{W_{N.C.f} = \Delta U}$

↳ only valid when $KE = \text{const.}$

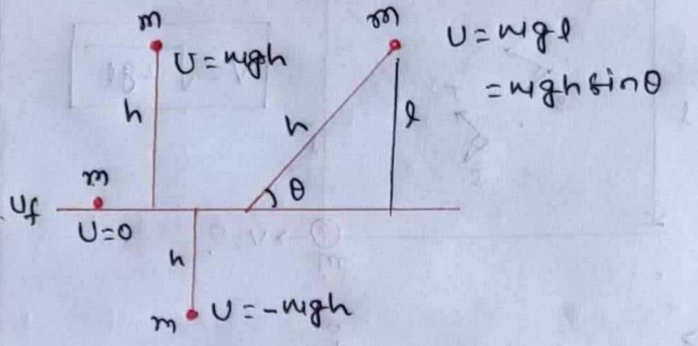
↳ N.C.f is acting against C.f.

हमारा force → N.C.f

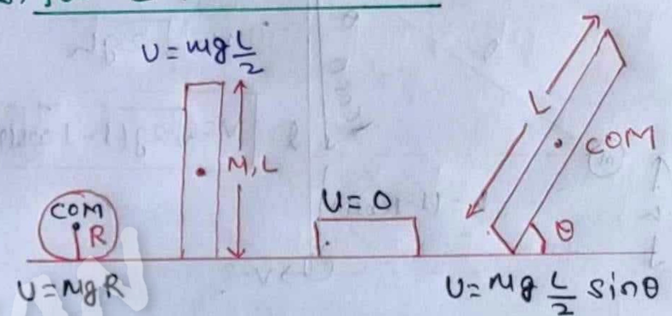
जहाँ C.O.M.E, वहाँ हम नहीं
जहाँ हम, वहाँ C.O.M.E नहीं

Gravitational potential Energy

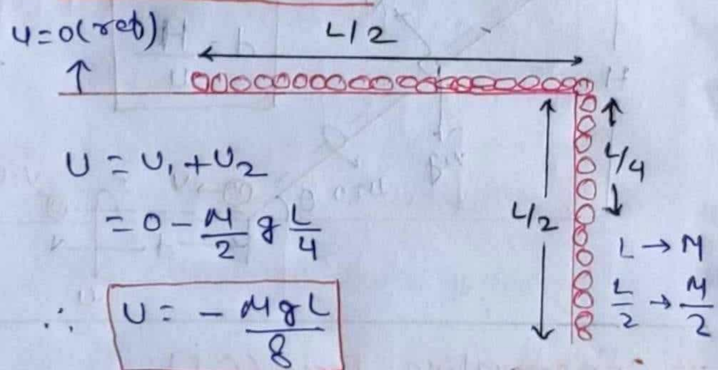
(A) For point Object



(B) For Extended object



Chain proplems

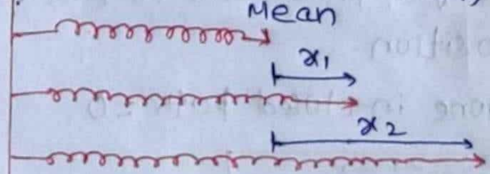


$U = U_1 + U_2$

$= 0 - \frac{M}{2} g \frac{L}{4}$

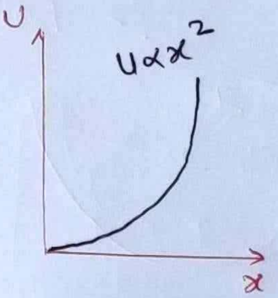
$\therefore \boxed{U = -\frac{MgL}{8}}$

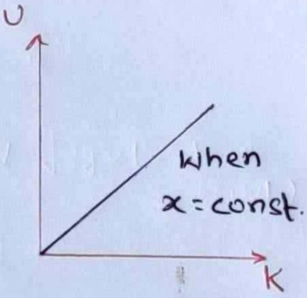
Spring Potential Energy



$\boxed{U_{x_2} - U_{x_1} = \frac{1}{2} k [x_2^2 - x_1^2]}$

If $x_1 = 0$; $U = \frac{1}{2} k x^2$

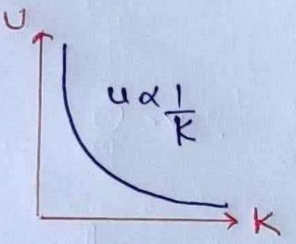




↳ $F = kx$

$U = \frac{1}{2}k\left(\frac{F}{K}\right)^2 = \frac{F^2}{2k}$

$\boxed{U = \frac{F^2}{2k}}$



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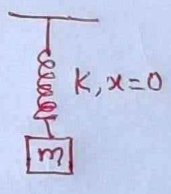


- Suddenly released

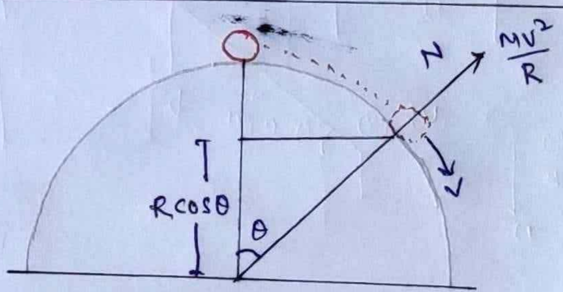
COME $\Rightarrow (KE+U)_i = (KE+U)_f$

$\Rightarrow 0+0+0 = 0 + \frac{1}{2}kx^2 - mgx$

$\Rightarrow \boxed{x = \frac{2mg}{K}} \rightarrow$ max elongation

- Slowly released

$\boxed{x = \frac{mg}{k}} \rightarrow$ max elongation



$V = \sqrt{2gR(1-\cos\theta)}$

$N + \frac{MV^2}{R} = Mg\cos\theta$

Lose contact → $N = 0$

$\frac{MV^2}{R} = Mg\cos\theta$

$\cos\theta = \frac{2gR(1-\cos\theta)}{gR} = 2 - 2\cos\theta$

$\cos\theta = 2/3$

$\boxed{\theta = \cos^{-1} 2/3}$

POWER

in time 't'

$P_{avg} = \frac{W_{Total}}{t_{total}}$

$= \frac{\Delta KE}{T}$

$= \frac{\int P.dt}{\int dt}$

at time 't'

$P_{inst} = \frac{dW}{dt}$

$= \frac{F ds}{dt}$

$= \vec{F}.\vec{V}$

- $(P_{avg})_{\text{by gravitation}} = \frac{MgH}{T}$
- Power → Scaler
  - Unit = watt = J/S
  - 1 HP = 746 watt = $\frac{3}{4}$ kW
- $P_{avg} = \frac{W_{Total}}{\text{Total time}} = \frac{\Delta KE}{T} = \frac{\int P.dt}{\int dt}$
- $P_{inst} = \frac{dW}{dt}$ = rate of work done

$= \vec{F}.\frac{d\vec{s}}{dt} = \vec{F}.\vec{V} = FV\cos\theta$

- If $\theta = 90°$, Power = 0 [θ → angle b/w F & V]
- If $\theta < 90°$, Power = +ve, Energy ↑
- If $\theta > 90°$, Power = −ve, Energy ↑
- If $\theta = 90°$, Power = 0

- Slope of W–T graph = Power
- Area of P–T graph = Work

- Efficiency, $\boxed{\eta = \frac{P_{out}}{P_{in}} = \frac{P_{work}}{P_{engine}}}$